DURHAM COLLEGE

COSC 1100

Class Exercise

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Class Exercise

Iteration: Designing a Program with Loops

Part 1 of 2

Iteration:
Designing a Program with Loops

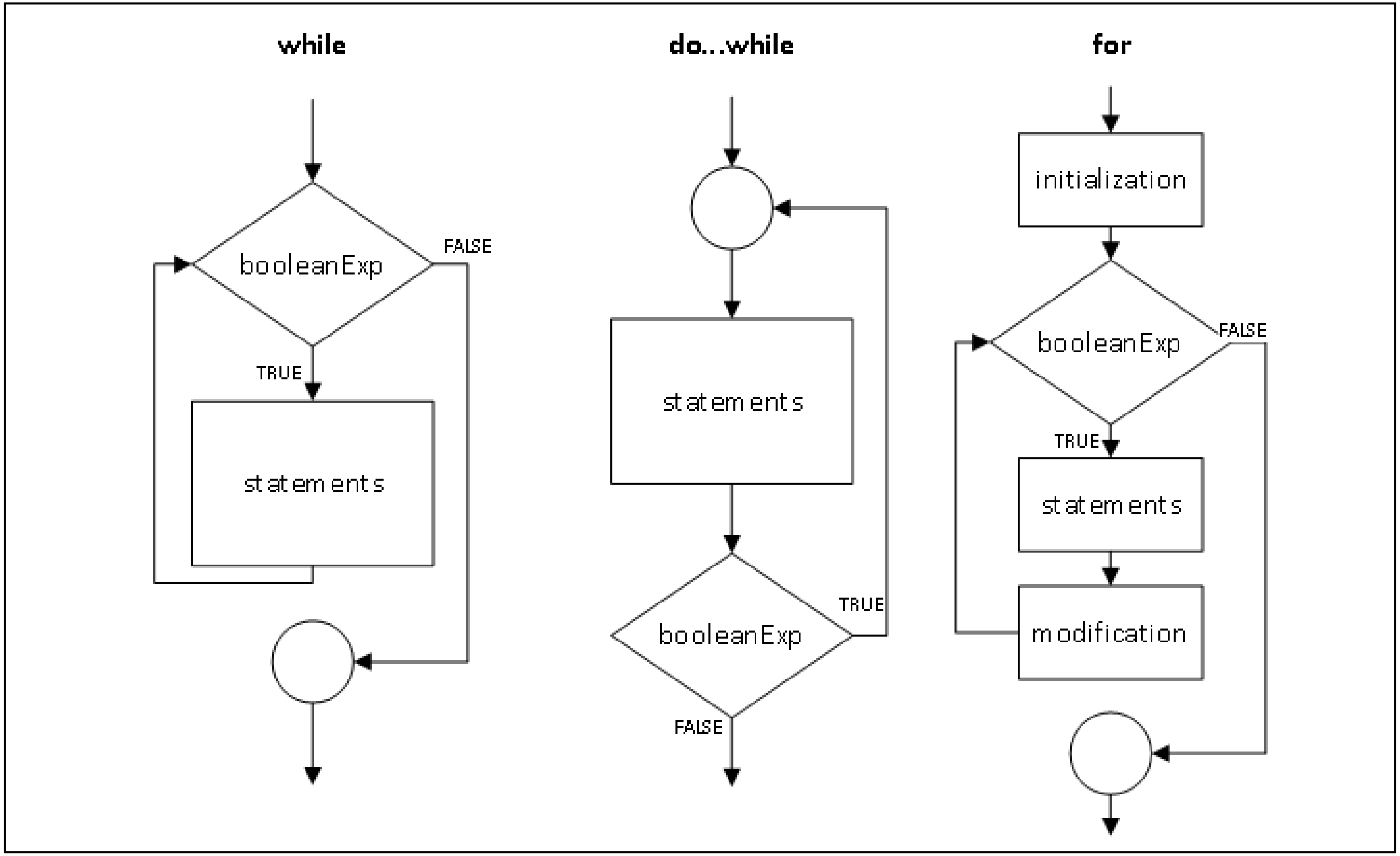
Goal of Part 1:
Determine the best way to use loops in a program's design.

Requirements

- Did you come prepared?
  - Ideally, watched COSC 1100 videos on While Loops and For Loops.

- Collaboration.
  - You will be encouraged to work together with 1-2 colleagues during this exercise. (i.e. groups of 2-3)
  - The group will share a flowchart based on their planning ideas at the end of this class, which will contribute to your grade.

Three Loop Structures



Why do we Use Loops?

- When should we use “for loops”?
- When should we use “while loops”?
- Why do we usually avoid break and continue?

Nested Loops for D&D Hit Points (Example)

```
# Loop to print column headings.
for column in range(1, 6):
  print(str(column), end="\t")
print("\n")
# Start of outer loop: a die size on each row.
for row in range(4, 12, 2):
  # Print the die size at the start of each row.
  print("\nd" + str(row), end="\t")
  # Start inner loop used for hit point values.
  for column in range(1, 6):
        # Determine and print the hit point values.
        print(str(int((row/2+1)*column+(row/2-1))), end="\t")
        # End of the inner loop (columns).
  # End of outer loop (rows).
```

Loops!

- Read the following problem statement, *for detailed comprehension*.
  1. Analyze the problem requirements and put your notes in a useful, sharable format.
     New this week: make a **flowchart** representing the entire algorithm.
  2. Make a discussion post including your notes / flowchart on DC Connect (**COSC 1100** -> **Activities** -> **Discussion** -> **Class Exercises**), including all group members' names.
  3. Be prepared to discuss your notes in class.

The Problem

John is the owner of a hotdog stand. He sells three (3) types of hot dogs, Traditional, Veggie Dog, and Curry Hot Dog. He would like a simple console based system to assist him in tallying and determining which type of hot dog is more popular.

As he intends to enter the data as the orders come in, he would like the system to present the options of the hot dogs in a repeated loop, with an extra option to tally and exit the system. This will be done with menu options like: 1, 2, 3, 4.

When an option is selected, the user is presented with a chance to enter a whole number, indicating the number sold.

The tally option (4) would display the number sold for each type of hot dog with a percentage of the total number of hot dogs sold before exiting.

Remember, John may not always hit the proper keys!

Class Exercise Part 1 Discussion

- Let’s review and compare some of the plans!

COSC 1100
Class Exercise

Iteration: Coding a Program with Loops

Part 2 of 2

Iteration:
Coding a Program with Loops

Goal of Part 2:

Writing a program with loops.

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Requirements

- Did you come prepared?
  - Ideally, watched COSC 1100 videos on While Loops and For Loops.
- Collaboration.
  - You may continue to work together with 1-2 colleagues during this exercise. (i.e. groups of 2-3)
  - Each group will submit their finished application for marks.

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The Problem (Review)

John is the owner of a hotdog stand. He sells three (3) types of hot dogs, Traditional, Veggie Dog, and Curry Hot Dog. He would like a simple console based system to assist him in tallying and determining which type of hot dog is more popular.

As he intends to enter the data as the orders come in, he would like the system to present the options of the hot dogs in a repeated loop, with an extra option to tally and exit the system. This will be done with menu options like: 1, 2, 3, 4.

When an option is selected, the user is presented with a chance to enter a whole number, indicating the number sold.

The tally option (4) would display the number sold for each type of hot dog with a percentage of the total number of hot dogs sold before exiting.

Remember, John may not always hit the proper keys!

Finish the Program!

1. Code your solution to the given problem based on planning notes/flowchart or one that was provided.
2. Chat, ask lots of questions, and don't be afraid of mistakes.
3. Either have it checked during class,
   or submit your group's finished Python code file to DC Connect
   (**COSC 1100** -> **Activities** -> **Assignments** -> **Class Exercise: Iteration**),
   including all group members' names.

This is meant to be done during class time; submissions received after our class time ends may be considered late.

Class Exercise Part 2 Summary and Discussion

- Loops are powerful, but they are also computationally expensive.

- Why do you think loops are expensive?